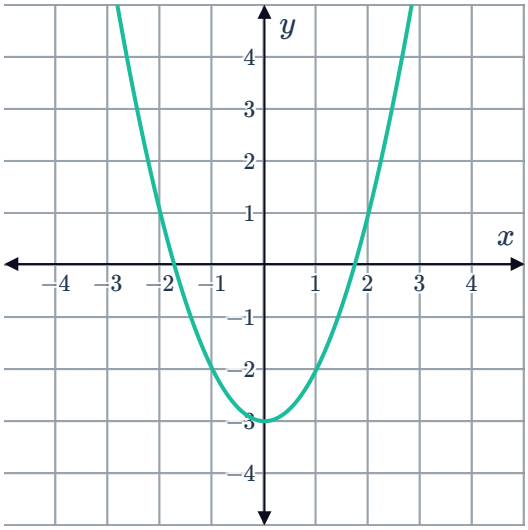
Translations

1

a

x	-2	-1	0	1	2
y	1	-2	-3	-2	1

b



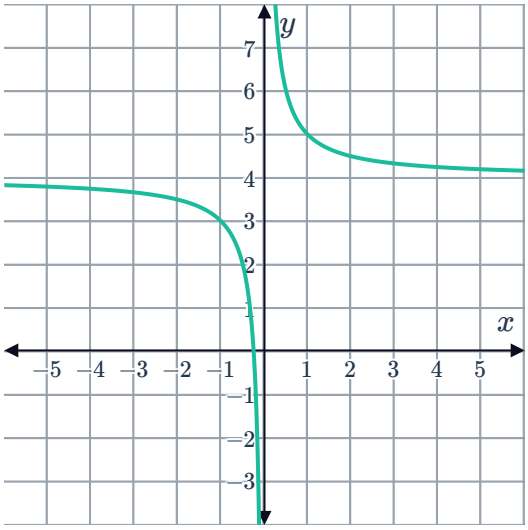
c $y = -3$

d Vertical translation

2

a Move the graph upwards by 4 units.

b

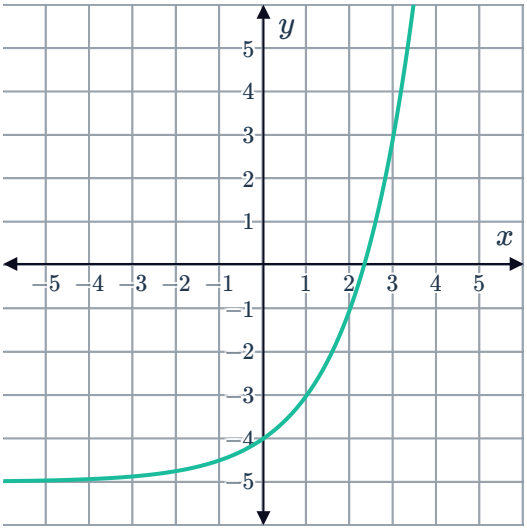


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

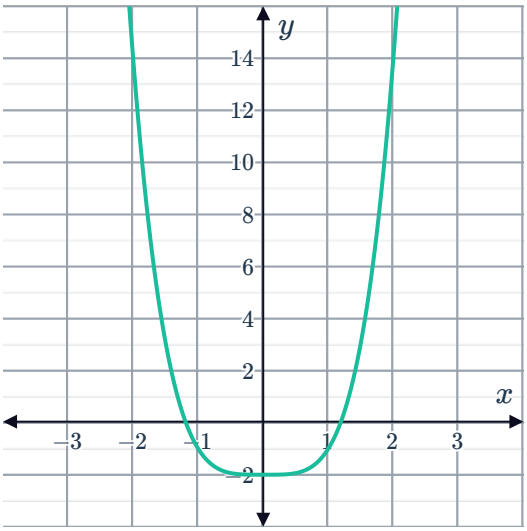
3

a Move the graph downwards by 5 units.

b



4



5

a Translate the graph up by 4 units.

b Translate the graph to the left by 6 units.

6

a $k = -4$

b Horizontal translation of 4 units

7 Move the graph to the left by 5 units.

8

a $y = (x - 10)^3$

b $y = 2^x - 9$

9

a $y = \dfrac{2}{x} + 2$

b $y = \dfrac{2}{x - 9}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

a $y = \sqrt{4 - x^2} - 7$

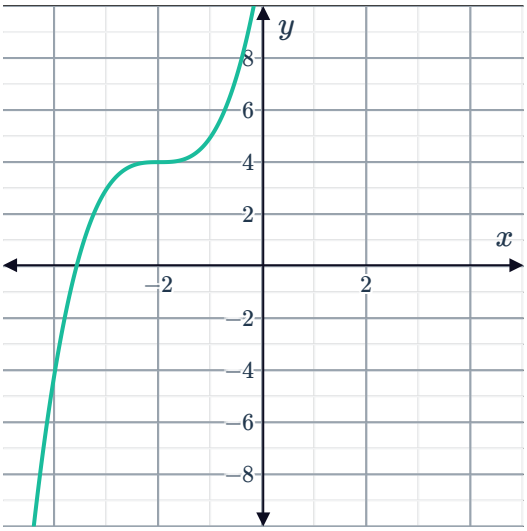
b $y = \sqrt{4 - (x + 3)^2}$

11 $y = (x - 8)^4 + 6$

12

a Move the graph to the left by 2 units and up by 4 units.

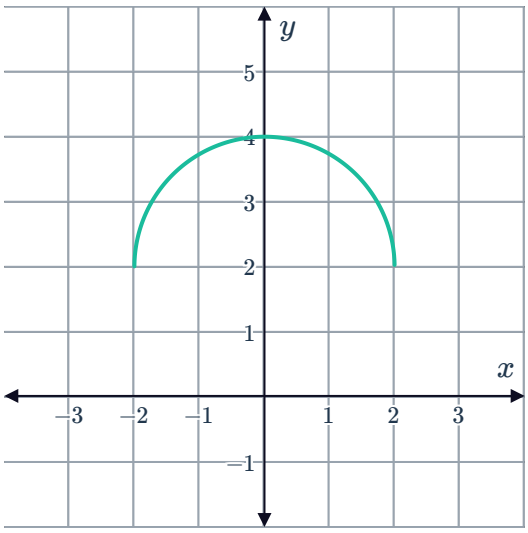
b



13

a Move the graph upwards by 2 units.

b

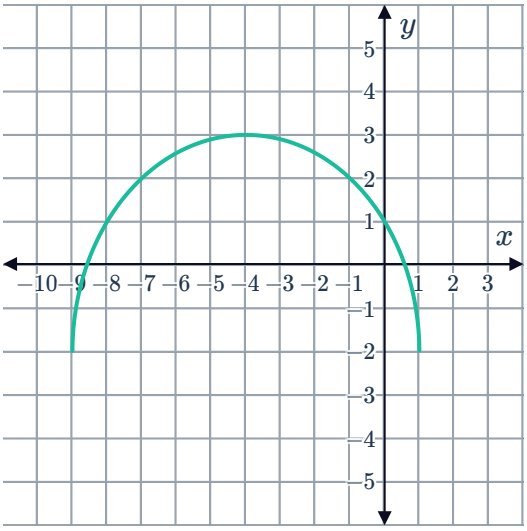


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

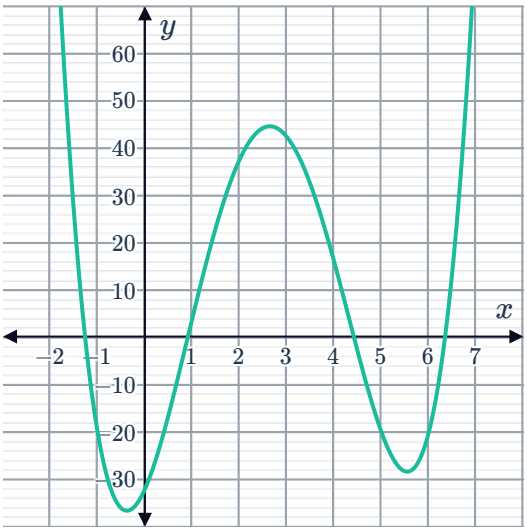
14

a Move the graph to the left by 4 units and downwards by 2 units.

b



15

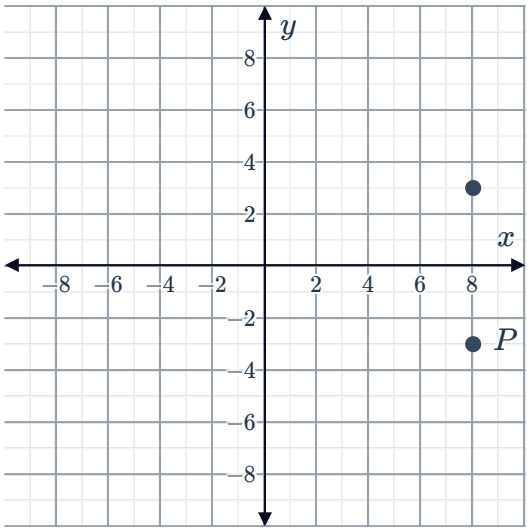


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

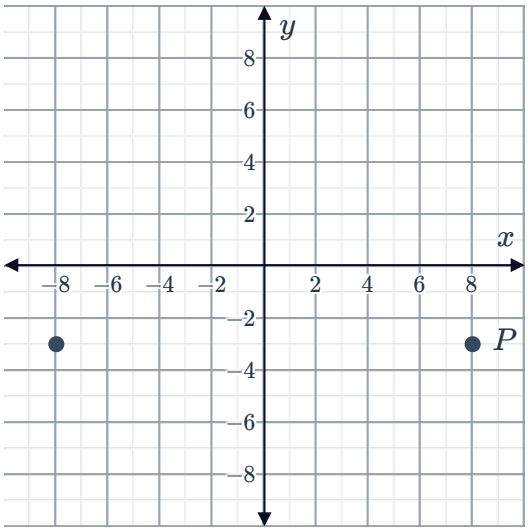
Reflections

16

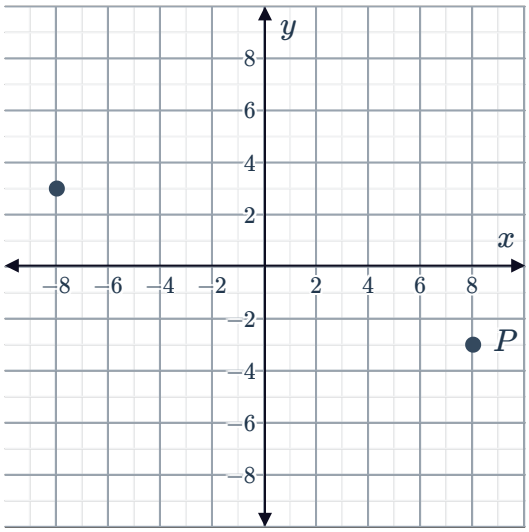
a



b



c



17

a Yes, the y-axis

b No

18 x-axis

19 No

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

20

a

i $f(-x) = -2x^3 + 2x$ ii The x-axis

b

i $f(-x) = 0.9x^6 - 7x^2 + 4$ ii The y-axis

c

i $f(-x) = -x^3 - x - 3$ ii Neither

d

i $f(-x) = \dfrac{1}{6x^3}$ ii The x-axis

21

a $(-6, -1)$ **b** $(6, 1)$ **c** $(6, -1)$

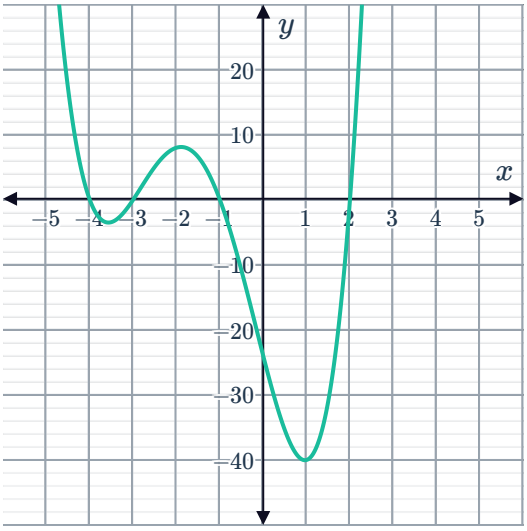
22 The graph is symmetric with respect to the x-axis, the y-axis, and the origin.

23

a $(-3, -2)$ **b** $(-3, 2)$ **c** $(9, 2)$ **d** $(-3, 2)$

e $(-3, -2)$

24



Dilations

25 $g(x) = \dfrac{1}{3}\sqrt{2x}$

26 $f\left(\dfrac{x}{k}\right)$ will be compressed horizontally and will appear narrower than $f(x)$

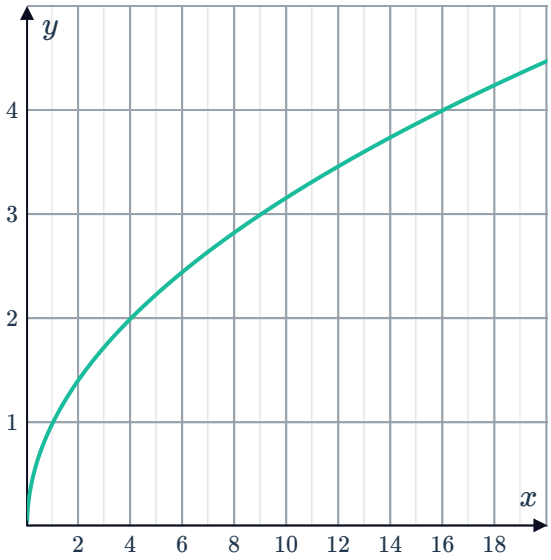
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

27

a

x	0	1	4	9	16
$f(x)$	0	1	2	3	4

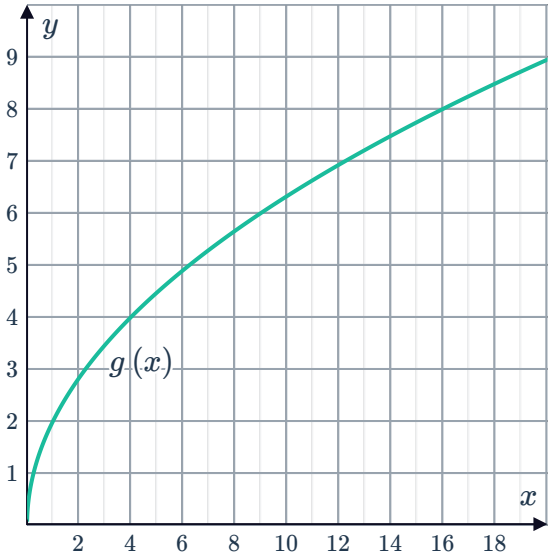
b



c

x	0	1	4	9	16
$f(x)$	0	1	2	3	4
$g(x)$	0	2	4	6	8

d

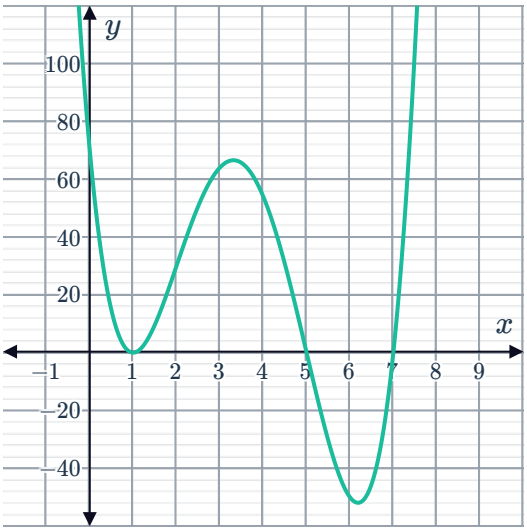


e The graph of $g(x)$ can be considered as the graph of $f(x)$ scaled vertically or horizontally.

f The square root of negative numbers is undefined.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

28



Combinations of transformations

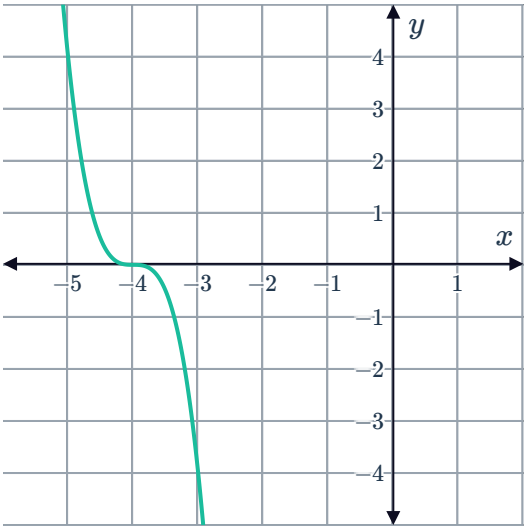
29

a $x = -9, 2$ **b** $x = -1, 10$ **c** $x = -5, 6$ **d** $x = 5, -6$

30

a $f(x) - 1$ **b** $f(4x)$ **c** $3f(x)$

31



32

a The function is translated 3 units left, undergoes a change in dilation of 3, is reflected about the horizontal axis and is translated 2 units up.

b $y = -3(x+3)^3 + 2$

33

a $g(x) = f(x) - 3$ **b** $g(x) = \frac{3}{x} - 3$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

34

a $g(x) = f(x+4)$

b $g(x) = 2^{x+4}$

35

a $f(x)$ is translated 3 units left and 5 units down and undergoes a change in dilation of factor 4 from the y-axis.

b $g(x)$ is translated 3units right and 3 units up and undergoes a change in dilation of factor $\frac{1}{2}$ towards the x-axis.

c $f(x)$ is translated 3 units down and undergoes a vertical dilation of factor 2 from the x-axis and horizontal dilation of factor 2 from the y-axis.

d $h(x)$ is translated 3 units up and undergoes change in dilation of factor 2 from the x-axis.

36

a $f(x)$ is translated 4 units left, undergoes a change in dilation of 7 and is reflected about the horizontal axis.

b $f(x)$ undergoes a change in dilation of 5, is reflected about the horizontal axis and is translated 8 units up.

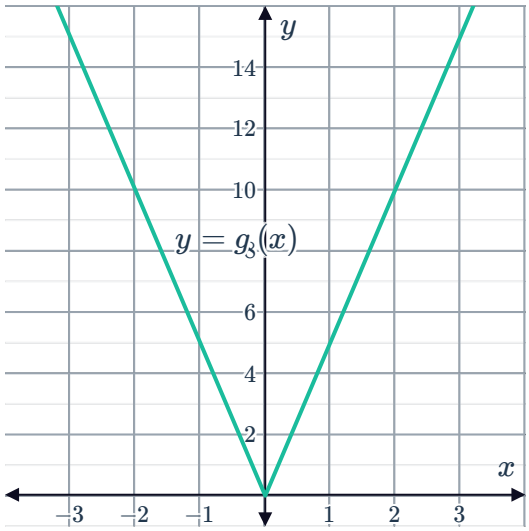
c $f(x)$ is translated 8 units left, undergoes a change in dilation of 10, is reflected about the horizontal axis and is translated 9 units down.

37

a

x	-3	-2	-1	0	1	2	3
$y = f(x)$	3	2	1	0	1	2	3
$g(x) = 5f(x)$	15	10	5	0	5	10	15
$h(x) = -2f(x)$	-6	-4	-2	0	-2	-4	-6

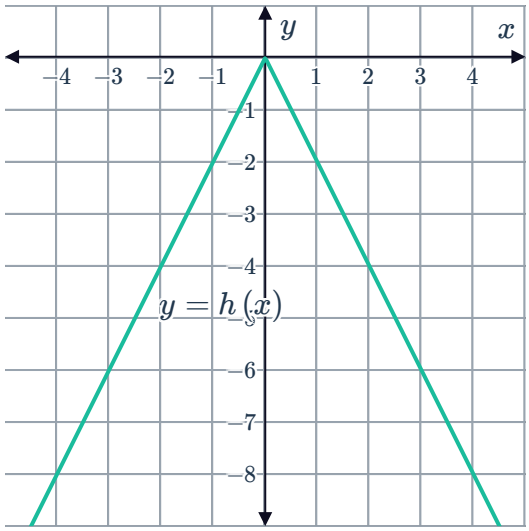
b



c Vertically stretched by a factor of 5

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

d



e Vertically stretched by a factor of 2 and reflected about the x-axis

38

a $f(x) = x^2$ **b** $g(x) = \frac{1}{4}f(x)$ **c** $g(x) = \frac{1}{4}x^2$ **d** $h(x) = -f(x)$

e $h(x) = -x^2$

39

Original point	$g(x)$	Corresponding point
$(9, -12)$	$g(x) = f(x) - 8$	$(9, -20)$
$(9, -12)$	$g(x) = 6f(x)$	$(9, -72)$
$(6, -7)$	$g(x) = f(x - 5)$	$(11, -7)$

40

a $(-11, -3)$ **b** $(0, -12)$ **c** $\left(-\frac{5}{6}, 3\right)$

41

Point	Transformation	Transformed point
$(-3, -1)$	$y = f(x - 5) - 2$	$(2, -3)$
$(0, 3)$	$y = \frac{1}{5}f(x - 3)$	$(3, \frac{3}{5})$
$(1, -2)$	$y = f(3x) - 2$	$(\frac{1}{3}, -4)$
$(-3, 5)$	$y = f(\frac{x}{2}) - 5$	$(-6, 0)$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647